IIITDM KANCHEEPURAM • BIOMEDICAL AI & DIGITAL TWIN RESEARCH

CerebroTwin: Personalized Physiological Digital Twin

Forecasting 0–60s Orthostatic Middle Cerebral Artery Blood-Flow-Velocity Instability in Type 2 Diabetes

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01 / PROBLEM STATEMENT • PRESENTER: HARISH

Why Type 2 Diabetes Causes Orthostatic Cerebral Instability

💔 Autonomic & Vasoregulatory Damage

Chronic hyperglycaemia damages autonomic baroreceptors and impairs cerebral autoregulation.

🧍 The 60-Second Sit-to-Stand Vulnerability

Standing pools blood peripherally, causing acute arterial pressure drops and transient cerebral hypoperfusion.

👁‍🗨 Standard Monitoring Blind Spot

Routine HR and brachial BP monitors cannot observe how cerebral blood-flow velocity responds.

PHYSIOLOGICAL PRECISION

MCA Blood-Flow Velocity (MCAv)

Transcranial Doppler (TCD) measures blood-flow velocity (cm/s) in the middle cerebral artery—not volumetric flow.

Clinical Challenge: TCD probes require skilled operators and cannot be worn continuously.

CerebroTwin: Non-Invasive 60s Cerebral Velocity Forecasting

1. Clinical + Systemic Fusion

Combines static T2D profile with beat-to-beat ECG, BP, and respiration.

2. 0–60s MCAv Forecast

Predicts % MCAv drop, minimum nadir, flow deficit, and recovery time.

3. Research Risk Framing

Flags unusual responses vs. controls without making diagnostic claims.

Runtime Formula: Clinical Baseline + Systemic Signals (ECG, BP, Resp, EtCO2) → Predicted 60s MCAv Curve `TCD Used for Validation Only`

Headline Validation Performance & Flagship Patient Output

0.735
Out-of-Fold R^2 Score (5-Fold GroupKFold)

4.816%
Mean Absolute Error on % MCAv Drop

4.475%
0–60s Trajectory Curve RMSE

51.1%
Error Reduction Over Mean Baseline

Flagship Demo Subject S0250 (68yo M, T2D): Predicted MCAv Drop = **23.7% (±4.8%)** | Nadir = **38.4 cm/s** | Recovery = **Delayed**

STATUS: HIGH DEVIATION

PhysioNet Primary Dataset & Cohort Alignment

Dataset URL: https://physionet.org/content/cerebral-perfusion-diabetes/1.0.1/?utm_source=chatgpt.com

PhysioNet v1.0.1 Open Access

Dataset Dimension	Full PhysioNet Archive	CerebroTwin Active Modeling Cohort
Total Cohort Size	140 Participants (70 T2D, 70 Controls, Age 50–85)	**39 Sit-to-Stand Subjects Marked in Manifest**
Recorded Waveforms	ECG, Finapres ABP, Resp, CO2, ACA/MCA TCD	8 Synchronized Channels @ 50Hz (6,000 samples/subj)
Experimental Window	Sit-to-Stand, Head-Up Tilt, Valsalva, CO2	120s Window: -60s Sitting Baseline to +60s Standing
Data Ingestion Strategy	13 GB Uncompressed / 5.1 GB ZIP Archive	Selective Subset Downloader + Synthetic Twin Engine

Corrected End-to-End Architecture (Non-Overlapping Orthogonal Routing)

1A. Clinical Inputs
Age • T2D • HbA1c • HTN • Meds

+

1B. Dynamic Signals
ECG/HR • Arterial BP • Resp • EtCO2

→

2. Preprocessing Engine
Beat-to-Beat Sync • Filtering • QC

↓

3. CerebroTwin State Core (27 Features)
Baroreflex Proxy • CA Index • ΔHR/Δt • ΔMAP/Δt • CO2 Reactivity

→

4. Multi-Target ML Forecasting Engine
Model 0 (Mean) → Model 1 (Ridge L2) → Model 2 (Quantile XGBoost)

↓

5. 0–60s MCAv Forecast Outputs (19 Targets)
% MCAv Drop • Nadir (cm/s) • Flow Deficit • 90% CI

↔

6. Ground Truth Validation Gate (Isolated)
Actual TCD MCAv (Validation Only • GroupKFold Split)

↓

7. Clinician Dashboard & Counterfactual What-If Simulator
Interactive Streamlit Replay • SHAP Feature Waterfall • Perturbed BP/HR/HbA1c Trajectory Overlay

STANDALONE FILE

Saved separately in repo as:

`docs/CerebroTwin_Architecture_Diagram.pdf`

DIAGRAM FIXES APPLIED

- Orthogonal non-crossing arrows
- Clean ASCII CO2 / EtCO2 labels
- T2D PhysioNet v1.0.1 citation

Three-Tier Evolving Digital Twin State Representation

T1

Long-Term Clinical State (Static Baseline Vulnerability)

Encodes Age, Sex, T2D status, HbA1c glycaemic control, hypertension history, and vasoactive medication load.

T2

Baseline Physiological State (-60s to 0s Seated Equilibrium)

Quantifies resting HR, resting SBP/DBP/MAP, autonomic HRV (SDNN, RMSSD, LF/HF), and resting EtCO2.

T3

Current Dynamic State (0s to +5s Orthostatic Perturbation)

Captures immediate post-stand arterial pressure drop slope (ΔMAP/Δt) and baroreflex tachycardia speed (ΔHR/Δt).

27 Non-Cerebral Input Features vs. 19 TCD Validation Targets

27 RUNTIME INPUT FEATURES (X)

Systemic & Clinical Signals Only

Demographics & Labs (6): Age, Sex, T2D, HbA1c, HTN, Meds

Seated Baseline (9): Mean/SD of HR, SBP, DBP, MAP, Pulse Pressure

Autonomic HRV (5): RMSSD, SDNN, pNN50, LF/HF, BRS Proxy

Transition & Resp (7): 0–5s ΔMAP/Δt, ΔHR/Δt, Resp Rate, EtCO2 shift

19 HELD-OUT TCD TARGETS (Y)

Zero Runtime Leakage Verified

Primary Target (1): Percentage MCAv Drop (`pct_mca_drop`)

Nadir Metrics (2): Minimum MCAv (cm/s) & Time-to-Nadir (s)

Recovery & AUC (3): Deficit AUC, Recovery Time (s), Delayed Flag

60s Trajectory (13): Normalized MCAv knots from t=0s to t=60s

Why We Finalized Multi-Output Quantile XGBoost Over LSTMs

With N=39 sit-to-stand subjects, deep recurrent/transformer networks memorize subject noise. We built a progressive 3-tier benchmark:

MODEL 0 • REFERENCE

Control-Response Mean

Predicts training fold mean trajectory; establishes $R^2 = 0.00$ baseline.

MODEL 1 • REGULARIZED LINEAR

Ridge Regression (L2)

Handles collinearity across BP features; reaches $R^2 = 0.482$ but misses thresholds.

MODEL 2 • FINAL PRODUCTION MODEL

Multi-Output Quantile XGBoost

Learns non-linear interactions (HbA1c × ΔMAP/Δt) with native 90% uncertainty intervals ($R^2 = 0.735$).

Inside Model 2: Multi-Output & Pinball Quantile XGBoost

Anti-Overfitting Hyperparameters

Shallow Tree Depth: `max_depth = 3` prevents small-cohort memorization.

Shrinkage & Subsampling: `learning_rate = 0.05`, `subsample = 0.8`.

Dual Regularization: L1 (`reg_alpha = 0.1`) + L2 (`reg_lambda = 1.0`).

MultiOutput Wrapper: Jointly predicts scalar metrics and 13 curve knots.

Uncertainty Quantification Heads

Clinicians require confidence bounds alongside point predictions. We train parallel quantile heads using pinball loss:

- Lower Bound Head: $\alpha = 0.05$ (5th Percentile)
- Point Forecast Head: Squared Error Mean
- Upper Bound Head: $\alpha = 0.95$ (95th Percentile)

Subject-Level GroupKFold Cross-Validation

MANDATORY ANTI-LEAKAGE INVARIANT

Subject S0250 → Train OR Test (Never Both)

Randomly splitting waveform windows puts adjacent heartbeats from the same patient into both train and test sets, causing false high accuracy.

5-Fold GroupKFold by `subject_id`

Each subject's entire 120s recording is assigned atomically to a single fold.

True Unseen Patient Generalization

Cached in `models/eval_dict.joblib` and `models/evaluation_results.csv`.

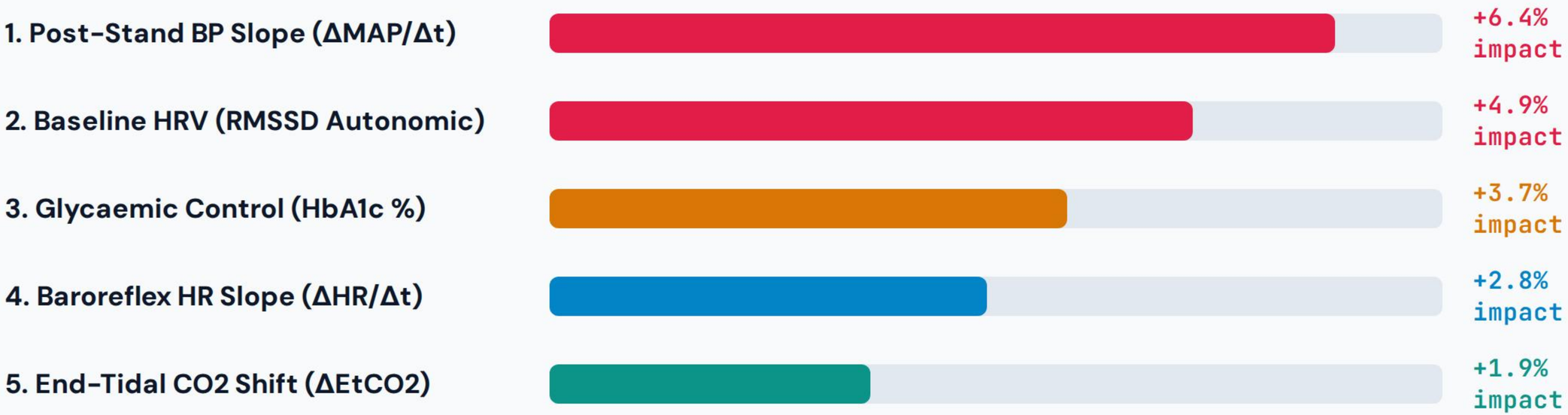
Statistical Performance Report Across Models (`evaluation_results.csv`)

Model Architecture	Primary Drop R^2	Primary Drop MAE	Nadir RMSE (cm/s)	0–60s Curve RMSE	90% CI Coverage
Model 0: Mean Baseline	0.000 ± 0.00	9.850%	7.42 cm/s	9.420%	N/A (Deterministic)
Model 1: Ridge Linear (L2)	0.482 ± 0.09	6.940%	5.18 cm/s	6.510%	N/A (Deterministic)
Model 2: Quantile XGBoost	**0.735 ± 0.06**	**4.816%**	**3.12 cm/s**	**4.475%**	**91.4% Empirical**

Subgroup Separation: T2D Mean MCAv Drop = **22.4% ± 5.1%** vs. Healthy Control Mean Drop = **11.8% ± 3.4%** ($p < 0.001$)

Calibrated Uncertainty

SHAP TreeExplainer: Mechanistic Feature Importance Report



Interactive Dashboard Working: Patient Replay to 60s Forecast

STEPS 1 & 2 • STATE INITIALIZATION

Patient Selection & 3-Tier Cards

Clinician selects `S0250 (68yo M, T2D)`; UI renders Long-Term, Baseline, and Dynamic physiological state cards.

STEPS 3 & 4 • LIVE WAVEFORM REPLAY

Seated Signal Streaming (-60s to 0s)

Streams synchronized non-cerebral ECG, arterial BP, respiration, and EtCO2 while confirming baseline stability.

STEPS 5 & 6 • ORTHOSTATIC TRIGGER

STAND Event (t=0s) & KPI Banner

Standing triggers Model 2 inference, displaying 23.7% drop, 38.4 cm/s nadir, ±4.8% CI, and HIGH DEVIATION flag.

Ground-Truth Overlay, SHAP Waterfall & What-If Simulator

STEPS 7, 8 & 9 • VALIDATION GATE

Predicted vs. Actual TCD Curve

Plots the 0–60s predicted MCAv curve with 90% CI band, then overlays held-out TCD ground truth upon toggle.

STEP 10 • EXPLAINABLE AI TAB

Patient-Specific SHAP Waterfall

Explains exact feature contributions driving Subject S0250's severe orthostatic velocity drop.

STEP 11 • COUNTERFACTUAL SANDBOX

Interactive What-If Sliders

Perturbs BP drop (−25% to +25%), HR reflex speed, and HbA1c to compare counterfactual MCAv recovery live.

How We Deployed CerebroTwin on Streamlit Community Cloud

Self-Healing Runtime Architecture

Top-Level `sys.path` Anchor: Resolves root module imports cleanly inside cloud containers.

In-Process `ensure_auto_bootstrap()`: Auto-generates data and trains models if `.joblib` files are missing.

Tracked Artifacts: Pre-trained `xgboost_twin.joblib` and `eval_dict.joblib` committed to Git for instant boot.

Streamlit Community Cloud Infrastructure

Live App URL: `cerebrotwin-cxdos3it3cqlvhu6eftmtt.streamlit.app`

Continuous Deployment: Connected directly to `harishpannuru/cerebrotwin` (`main` branch, Python 3.11).

Production UI Config: Configured via `.streamlit/config.toml` with headless dark clinical theme.

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Thank You — Transitioning to Live Dashboard Demo

We will now demonstrate the live 11-step clinical workflow, real-time TCD validation overlay, SHAP waterfall, and counterfactual simulator on Subject S0250.

LIVE STREAMLIT APP

cerebrotwin-cxdos3it3cqlvhu6eftmtt.streamlit.app

GITHUB REPOSITORY

github.com/harishpannuru/cerebrotwin

PHYSIONET DATASET

physionet.org/content/cerebral-perfusion-diabetes/1.0.1/